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Conformation Based in silico Studies of Cyclic Tetrapeptides with βγ Fused Turns as Thrombin Inhibitors

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The enormous social and economic consequences of thrombotic diseases, as well as the significant dangers associated with present medications, need the development of more effective and safer anticoagulants. *In silico* structure-based design was used to find new peptide-based thrombin inhibitors and cyclic peptide-based molecules are attracting wide attention in that regard. Here we have shown the synthesis, conformational and *in silico* analysis of tetrahydrofuran amino acid (TAA) based 15-membered cyclic tetrapeptides (CTPs) containing guanidine side chain as thrombin inhibitors. Solution NMR conformational analysis suggests that the βγ fused structures of CTPs are retained in the core and there is no effect of guanidine functionality on the scaffold. Molecular docking results showed that these CTPs can inhibit thrombin protein by interacting with Asp189 residue and the core structure helps to retain the guanidine group inside the active site of protein without any alteration in the backbone H-bonding. Overall, we have presented here the synthesis, conformational analysis of TAA containing cyclic tetrapeptides along with their docking study against thrombin protein. Further these results will help to design peptide based novel thrombin inhibitors.

Introduction

Peptides and peptide based inhibitors are emerging as a safer molecule for drug designing having their several advantages over the small molecules.[1] Peptides based drugs show high order of specificity for their biological targets and excellent cellular penetration.[2] During the last two decades, several efforts have been made to develop novel peptide based drugs.[3] Currently more than 200 peptide based drugs are evaluated for the clinical trial and 7000 naturally occurring peptides have been identified as a biologically active molecule.[4] Considering the fact that peptides are the foremost choice for the drug development still several developments are required to increase their efficacies and therapeutic profiles. Naturally occurring peptides having some drawbacks and not suitable for the convenient druggable molecules as they have poor metabolic profile, low oral availability and also poor membrane permeability.[5]

Several cyclic peptides are reported which are showing excellent biological activities. Among them, the cyclic tetrapeptides (CTPs), having different types of turn structures, are emerging as the privilege class of molecules as protein protein Inhibitors (PPi).[6] Most of CTPs adopted rigid structure which is also favorable for the epitope design.[7] In addition, CTPs are resistant to the cellular proteolytic enzymes.[8] In continuation of our decades' long ongoing studies on sugar amino acid containing peptidomimetics,[9] we also synthesized tetrahydrofuran amino acid (TAA) based CTPs (Figure 1) by modifying the stereocenters on the '2,5' positions of the tetrahydrofuran ring. These cyclopeptides were highly rigid and adopted the βγ-fused turn structures.[10] We investigated these structures by extensive NMR studies and our results showed that these fused structures were stable and peptide core structure was also not altered after stereocenter modification on the tetrahydrofuran ring. These CTPs showed excellent HDAC class-1 inhibitory activities.[10] We used the 15-membered fused core as a model peptide to design other inhibitors and modified the side chain by inserting the guanidine group as a potential thrombin inhibitor. Guanidine is a well-known pharmacophore having thrombin inhibition activity by modifying the thrombin active site.[11]

Thrombin protein regulates clot formation by converting the fibrinogen to fibrin.[12] In the cellular system, thrombin protein forms prothrombin via coagulation cascades. Thrombin

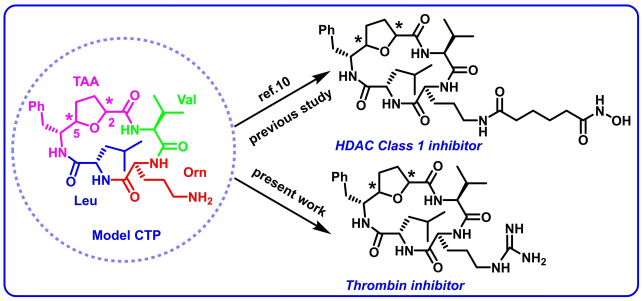


Figure 1. Structure of 15-membered cyclic peptides with βγ-fused turn structures: A) ornithine attached with hydroximic acid as a HDAC class 1 inhibitor; B) ornithine converted to guanidine as a potential thrombin inhibitor.

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is a serine protease regulating many signaling cascades associated with blood coagulation.[13] However, in some instances where its level increases there arise some pathological conditions like hemorrhage, stroke, and neurodegenerative diseases like traumatic brain injury and primary CNS tumors.[14] Structurally thrombin protein has two parts, A and B chains. Chain A and B contain 36 and 259 amino acids, respectively and both the chains are connected by disulfide bridges having three active sites S1, S2, and S3.[15] The catalytic triad has some critical amino acids Asp189, Leu99 and asp102 which is the binding interface for the S1, S2 and S3.[16] So in order to discover selective and direct thrombin inhibitors, the effort started with a tripeptide template, D–Phe–Pro–Arg–H.[17] This fragment interacts with three essential binding sites on the thrombin enzyme, which include the S1 specific pocket and two hydrophobic pockets, the proximal S2 and distal S3. Asp189 in the S1 specific pocket forms a salt bridge with the guanidine functionality. In order to take an advantage of this interaction, many initial inhibitors usually contained a guanidine or other highly basic groups, such as benzamidine and imidazole. This strategy often provides molecules with increased potency.

Several Inhibitors have been reported as thrombin inhibitors like, for example, melagatran, dabigatran etexilate, and argatroban.[18] Melagatran is one of the potent antithrombin molecules having a side-chain guanidine group that interacts with the Asp189 residue in the S1 pocket of the protein.[19] Cyclotheonamide A, a naturally occurring macrocyclic pentapeptide with arginine as one of the residues was reported as thrombin inhibitor.[20] A library of designed cyclic peptides with guanidine functionality has been reported by Heinis group as human thrombin inhibitors.[21] Based on that concept and the model cyclic peptides reported by our group (Figure 1),[10,22] we have designed tetrahydrofuran amino acid containing cyclic tetrapeptides **1 A**–**B** and **2 A**–**B** with guanidine functionality in the side chain (Figure 2). These cyclic tetrapeptides were derived from our previously reported model cyclic peptides.[22]

Asp189 in thrombin is a critical residue and regulates its signaling pathways.[23] It interacts with the sodium ions and coordinates the interaction with the substrate arginine residue. In addition, replacement of the Asp189 with alanine, glutamic acid and serine, suggested the functional loss of the thrombin protein. So we designed CTPs in a way so that the cyclic core can interact with the hydrophobic pocket of thrombin protein (S2 & S3) and the Arg side-chain can insert the guanidine group in the pocket of S1 to interact with Asp189 (Figure 3). Here we have done conformational study of our CTPs and *in silico* study with thrombin protein to see their nature of binding with the protein.

Results and Discussion

Synthesis: The CTPs **1 A**–**B** were synthesized from our reported cyclic peptides through conversion of ornithine to arginine. The basic cyclic core of peptides **1 A**–**B**/**2 A**–**B** were synthesized by cyclization of H_2N–Taa–Val–Orn(Z)–Leu–OH using FDPP in dry DMF under high dilution following our earlier reported procedures.[22c–d] Cbz-deprotection of the resulting cyclic peptide, *cyclo*-[Taa–Val–Orn(Z)–Leu] resulted into **3** and **4**. Compounds **3** and **4** were allowed to react with *N*,*N*′-bis-Boc-*S*-methylisothiourea[24] in the presence of $HgCl_2$ in DMF to obtain compounds **1 A** and **1 B**, respectively in 63/65 % yields. On treatment with TFA in CH_2Cl_2, cyclic tetrapeptides *cyclo*-[Taa–Val–Arg(Boc_2)–Leu] (**1 A** and **1 B**) resulted the free arginine based cyclicpeptide analogs **2 A** and **2 B**, respectively in 71/76 % yields (Scheme 1).

Conformational and Molecular Dynamics (MD) studies: Detailed solution conformational studies were carried out for cyclic tetrapeptides (CTPs) in DMSO-d_6 with 2–10 mM concentrations to investigate their structural and conformational preferences in the solution. Sequential assignment of all proton resonances was performed as previously described.[10] All cyclic tetrapeptides (CTPs) are similar in the backbone, however different on the side chain protective groups of Orn reside. All the amide protons were resonating between δ 6.0 ppm to δ 8.5 ppm scale. All the proton spectra depicted well-dispersed and a single set of resonances in the NMR time scale indicating that the peptides existed in stable single conformations. For the NMR assignments, a series of 2D NMR experiments were carried out and all the assigned details are listed in the table

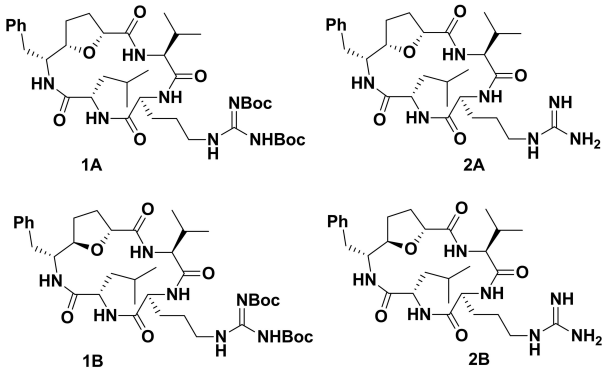


Figure 2. General structures of the proposed cyclic tetrapeptides **1**, **2** having guanidine side chain.

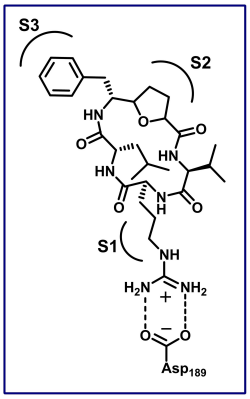
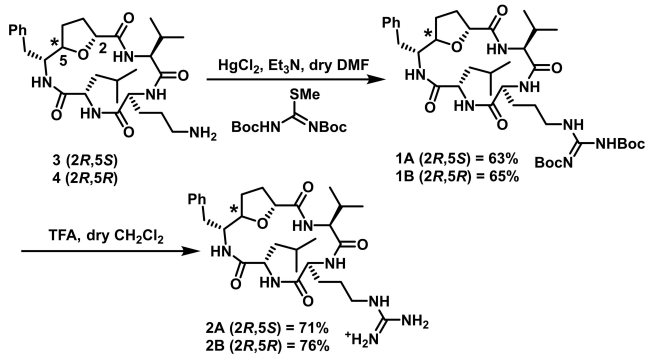


Figure 3. Schematic representation of the binding mode of our designed CTPs to the active site of thrombin.

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Scheme 1. Synthesis of Taa containing cyclic tetrapeptides **1 A**–**B**/**2 A**–**B**.

(see the Supporting Information for all the CTPs. The nOe data were extracted from the 2D-ROESY experiment (Figure 4).

The nOe intensities were converted to distances as described earlier.[10] The detailed protocol for the MD simulation is listed in the experimental section in the Supporting Information. The MD studies showed results similar to what was observed in our previous study where we used this fused turn system to design a HDAC inhibitor. Based on MD studies, 10 best superimposed structures of CTPs are presented here (Figure 5).

Docking studies with thrombin protein: BIOVIA Discovery Studio Visualizer (https://discover.3ds.com/discovery-studio-visualizer-download) was used for the editing/minimizing the 3D structures of cyclic **2 A** and **2 B**. All the structural coordinates were checked and further saved in PDB format. We used

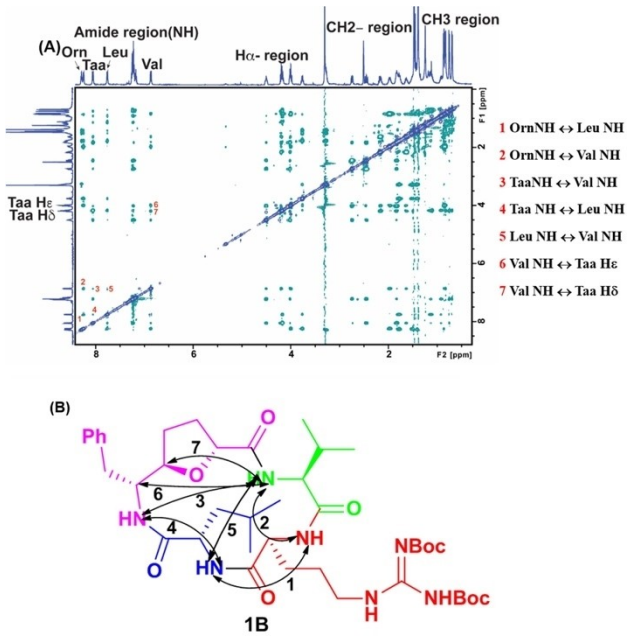


Figure 4. A) Important NOEs of cyclic tetrapeptdepeptide **1 B**, where amide hydrogen and other hydrogen are showing the correlations. B) The NOEs are represented on the structure and indicated by the solid black arrows (1–7).

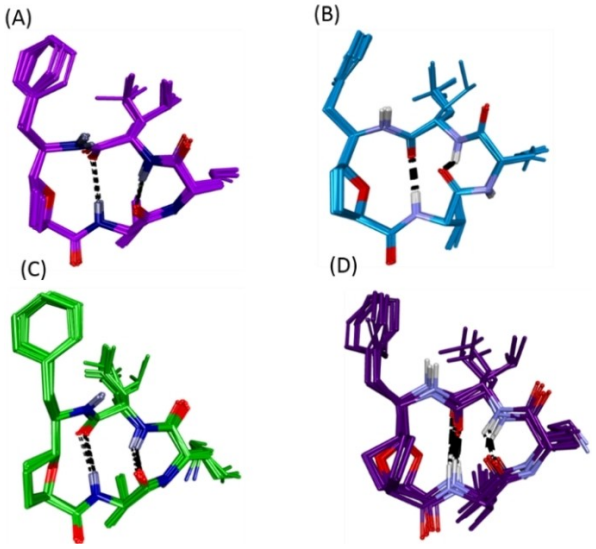


Figure 5. Solution NMR ensemble structures of cyclic peptides **1 A**, **1 B**, **2 A** and **2 B** (A to D, respectively); 10 best structures were superimposed and the calculated RMSD was observed <1 for all the peptides. Side chain hydrogens and the protected group are omitted for structural clarity.

AutoDock tools for the docking studies, using standard docking protocol. Input files for AutoDock4.2 were prepared using AutoDockTools and Autogrid4.2.

During molecular docking, peptidomimetics were considered as a flexible and Gasteiger partial charges were assigned on the atoms. 3D structure of human thrombin was separated from the crystal structure of thrombin in complex with double domain Kazal inhibitor rhodniin (PDB ID:1TBQ) and was used as a receptor protein during docking. Thrombin was considered as rigid molecule and Kollman partial charges were assigned on all atoms. Binding site was defined by a grid box enclosing Asp189 (box center at 15.57, −13.92 and 19.09 and having 50 points each in x, y and z directions, respectively). Lamarckian genetic algorithm was used to generate 50 conformations of each of **2 A** and **2 B**. All the docked poses were further analyzed by the visualizer software UCSF Chimera[25] and ADT docking.

Out of 50 docked conformations, two criteria were considered for the selection of probable binding mode of cyclic peptidomimetic **2 A**. Firstly, core of the cyclic peptidomimetic should not show high deviation, and secondly, the side chain of **2 A** bearing the guanidinium group should interact with Asp189. Selected docked conformation (Figure 6) fulfilled above criteria. In the S1 pocket of thrombin, Asp189 has been reported to be a very important residue which has a role in the substrate binding as well as in the monovalent cation specificity.[23] CTP **2 A** has shown a salt bridge and H-bond with Asp189 residue of thrombin, though **2 A** was also involved in formation of additional H-bond with G219. Apart from the salt bridge and H-bonds, several hydrophobic interactions and van der Waals interactions stabilize the binding of **2 A** (−5.21 kcal/mol) in the active site of thrombin. Comparison of docked conformation of **2 A** with a co-crystallized peptidomimetic (PDB ID: 1A5G)[26] showed similar placement of guanidinium group in

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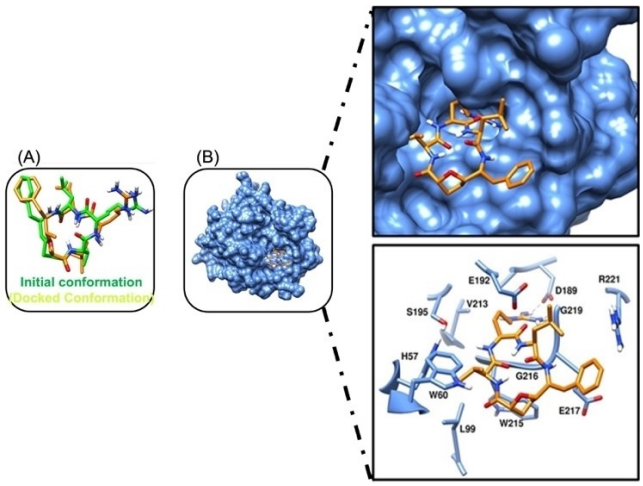


Figure 6. (A) Docked conformation of **2 A** aligned with the initial conformation and (B) Docked conformation of **2 A** in the active site of thrombin. Non-polar hydrogens have been omitted for the clarity.

the S1 pocket (Figure 7). In case of **1 A**, the guanidinium group is not available for the interaction with Asp189, leading to different binding pose (Figure 8) and weak interaction with thrombin (−4.75 kcal/mol).

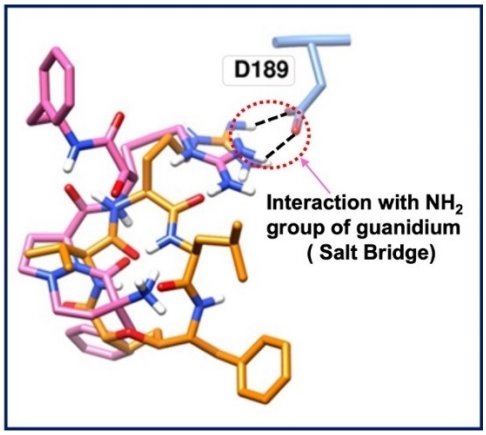


Figure 7. Comparison of the docked conformation of **2 A** (yellow) with the co-crystallized peptidomimetic and the interactions with Asp189 have been highlighted.

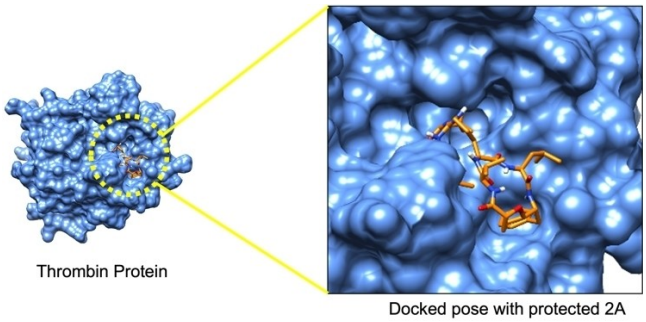


Figure 8. Docked conformation of **1 A** in the active site of thrombin.

Conclusion:

We explored the transformation of the side chain free amine of ornithin to guanidine functionality i.e. arginine residue. Like our previous reports, we have shown here that these 15-membered rings adopt a compact 10-membered and 7-membered hydrogen bonded βγ-fused turn structure. In addition, we also investigated the thrombin inhibitor properties of these cyclic peptides carrying the gunidine functionality in the side chain.

Guanidine group is expected to inhibit the thrombin activity by invading the interaction of the Asp189 residue. Structural and docking studies have shown that the side chain guanidine group of our TAA containing cyclic peptides bind with the S1 pocket of thrombin to potentially inhibit the Asp189 interaction. The amino moiety of guanidine group is primarily involved in the binding thus inhibiting the thrombin active site to any other substrate. In **1 A**, the protected guanidinium group did not show any interaction with Asp189. So, in conclusion we demonstrated here that these βγ-fused CTPs could serve as very useful building blocks for designing other inhibitors by side chain modifications as demonstrated here by Orn to Arg transformation giving rise to potential thrombin inhibitors.

Supporting Information Summary

General experimental details and instrumentation; detailed experimental procedures, spectroscopic data of the compounds; ^{1}HNMR and 2DNMR (TOCSY, ROESY) spectra of the compounds; details of Molecular Dynamics, and Molecular Docking studies.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Tetrahydrofuran amino acid · cyclic peptide · Thrombin · guanidine · βγ fused structure

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